

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. (Variation on NS 3.5.3) Use the determinant to find all values $\alpha \in \mathbb{C}$ such that the matrix

$$A = \begin{bmatrix} 1 & \alpha & 0 & 0 \\ \alpha & 1 & 0 & 0 \\ 0 & \alpha & 1 & 0 \\ 0 & 0 & \alpha & 1 \end{bmatrix}$$

is invertible.

2. (NS 3.6.11(a)) Let $A \in \mathbb{F}^{m \times n}$. Prove that the linear system $Ax = 0$ has a nonzero solution $x \in \mathbb{F}^{n \times 1}$ if and only if the n columns of A are linearly dependent.
3. Consider the three linear systems of equations:

$$\begin{array}{lll} x_1 + x_2 = -7 & x_1 + x_2 = 2 & x_1 + x_2 = 2 \\ x_2 + 2x_3 = 10 & x_2 + 2x_3 = 3 & x_2 + 2x_3 = 4 \\ 2x_2 + x_3 = 2 & 2x_2 + x_3 = 3 & 2x_2 + x_3 = -1 \end{array}$$

corresponding to the systems $Ax = b$, $Ax = c$, and $Ax = d$ respectively. Solve all three systems simultaneously by applying Gauss-Jordan elimination to the 3×6 augmented matrix $[A \mid b \ c \ d]$.

4. Consider the linear system of equations:

$$\begin{array}{l} x_1 - \alpha x_2 = 3 \\ x_2 - \beta x_3 = 0 \\ -\gamma x_1 + x_3 = -2 \end{array}$$

Find conditions on $\alpha, \beta, \gamma \in \mathbb{R}$ such that the system has no solutions, a unique solution, and infinitely many solutions. In the latter two cases, find the solution set. (*Note.* Your solution(s) may be in terms of α, β , and γ .)



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 3.5: 1, 4, 8, 10, 11
- Section 3.6: 1, 2, 4, 5, 11b, 12
- Section 3.7: 1, 2, 3, 4, 5, 7, 9
- Section 3.8: 31, 37, 38, 40

1. Let $A, B \in \mathbb{F}^{n \times n}$ and $\alpha \in \mathbb{F}$. Prove the following:

- (a) $\text{tr}(\alpha A) = \alpha \text{tr}(A)$
- (b) $\text{tr}(A^T) = \text{tr}(A)$ and $\text{tr}(A^*) = \overline{\text{tr}(A)}$.
- (c) $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$.
- (d) $\text{tr}(A^*A) = 0$ if and only if $\text{tr}(AA^*) = 0$ if and only if $A = 0$.

2. Show that for $A = [a_{ij}] \in \mathbb{F}^{2 \times 2}$, $\det(A) = a_{11}a_{22} - a_{12}a_{21}$.

3. Consider an invertible matrix $A = [a_{ij}] \in \mathbb{F}^{2 \times 2}$. Use Theorem 3.30 (14) to show that

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

4. Prove that a lower/upper triangular matrix is invertible if and only if all of the entries on its main diagonal are nonzero.

5. Let $A \in \mathbb{F}^{m \times n}$ and $b \in \mathbb{F}^{m \times 1}$. Prove that $\text{rank}(A) \leq \text{rank}([A \mid b])$. More specifically, prove that either $\text{rank}(A) = \text{rank}([A \mid b])$ or $\text{rank}(A) = \text{rank}([A \mid b]) - 1$.

6. (More characterization of invertible matrices) Let $A \in \mathbb{F}^{n \times n}$

- (a) Prove that if $\text{rank}(A) = n$, then $Ax = 0$ has no nontrivial solution.
- (b) Prove that if $Ax = 0$ has no nontrivial solution, then $Ax = b$ has a unique solution for all $b \in \mathbb{F}^{n \times 1}$.
- (c) Prove that if $Ax = b$ has a unique solution for all $b \in \mathbb{F}^{n \times 1}$, then $Ax = b$ is consistent for all $b \in \mathbb{F}^{n \times 1}$.



- (d) Prove that if $Ax = b$ is consistent for all $b \in \mathbb{F}^{n \times 1}$, then $Ax = e_i$ has a solution for each standard basis vector $e_i \in \mathbb{F}^{n \times 1}$.
- (e) Prove that if $Ax = e_i$ has a solution for each standard basis vector $e_i \in \mathbb{F}^{n \times 1}$ then $\text{rank}(A) = n$.

